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REVIEW

molecular informatics

Chemical language models for molecular design

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Abstract

In drug discovery, chemical language models (CLMs) originating from natural language processing offer new opportunities for molecular design. CLMs have been developed using recurrent neural network (RNN) or transformer architectures. For the predictive performance of RNN-based encoder-decoder frameworks and transformers, attention mechanisms play a central role. Among others, emerging application areas for CLMs include constrained generative modeling and the prediction of chemical reactions or drug-target interactions. Since CLMs are applicable to any compound or target data that can be presented in a sequential format and tokenized, mappings of different types of sequences can be learned. For example, active compounds can be predicted from protein sequence motifs. Novel off-the-beat-path applications can also be considered. For example, analogue series from medicinal chemistry can be perceived and represented as chemical sequences and extended with new compounds using CLMs. Herein, methodological features of CLMs and different applications are discussed.

KEYWORDS
drug design, language models, recurrent neural networks, encoder-decoder frameworks, transformers, attention mechanisms

1 | INTRODUCTION

For molecular design, deep neural network models from natural language processing are increasingly used [1]. These models are derived to translate one sequence of characters into another (mostly compound SMILES strings) and are often termed chemical language models (CLMs) [2,3]. They must learn the chemical vocabulary and syntax used to represent molecules and, in addition, conditional probabilities for the occurrence of characters at a given position in sequences depending on the preceding characters. To learn textual representations, CLMs are usually pre-trained on large numbers of molecular strings. Subsequently, they are fine-tuned on smaller data sets to focus on compounds with desired properties (such as activity against a target protein). All input data including chemical structures, molecular properties, or others must be tokenized. Preferred computational architectures for CLMs include recurrent neural networks (RNNs) [4,5] and transformers [6]. RNN language models process sequential data. Therefore, these models typically consist of long short-term memory (LSTM) cells designed to prevent vanishing gradients in RNN architectures [7]. Such RNNs can also be used as encoder and decoder modules of a variational autoencoder (VAE) with latent space [8,9]. In a VAE, the encoder compresses a sequence into a context vector in its last hidden state, providing the input for the decoder. To control potential errors when compressing long input sequences, a layer with attention (importance) mechanism can be added that operates not only on the last but all hidden states of the encoder and assigns importance weights to selected tokens of the input sequence [10]. In an attention-based VAE, this information is then used to decode the output sequence.

While RNNs were originally adapted as CLMs, and continue to be used for a variety of applications, there is increasing interest in transformer models that are described in more detail in the following section.

2 | TRANSFORMERS

Transformer networks further extend attention mechanisms. A distinguishing feature is that transformers do not contain RNN components but exclusively consist of multiple attention-based encoder and decoder modules [6], each of which combines multi-head self-attention sub-layers with a fully connected feed-forward neural network sub-layer. The multi-head self-attention mechanism is a hallmark of transformers that distinguishes them from other encoder-decoder architectures. In a multi-head self-attention sub-layer, several attention functions act in concert to process different segments of the input sequence and produce multiple attention vectors. Importantly, each attention vector is an independently derived entity. Thus, attention vectors can be processed in parallel. For the derivation of conditional probabilities, transformers capture long-range dependencies between tokens through parallel instead of sequential processing. Thus, multi-head self-attention contributes to the versatility of transformers in handling and translating different types of sequences and the associated parallel processing of attention vectors contributes to training efficiency. These features also explain the increasing popularity of transformer models for machine translation tasks in chemistry. Figure 1 schematically illustrates the architecture of a transformer trained to map protein sequences to compound strings and predict novel active compounds from sequence motifs of binding sites (*vide infra*) [11]. Many CLMs consist of corresponding transformer architectures.

The encoder (left) consists of three modules each containing eight multi-head attention sub-layers and a feed forward sub-layer that generates a 512-dimensional vector of input data through positional encoding. The

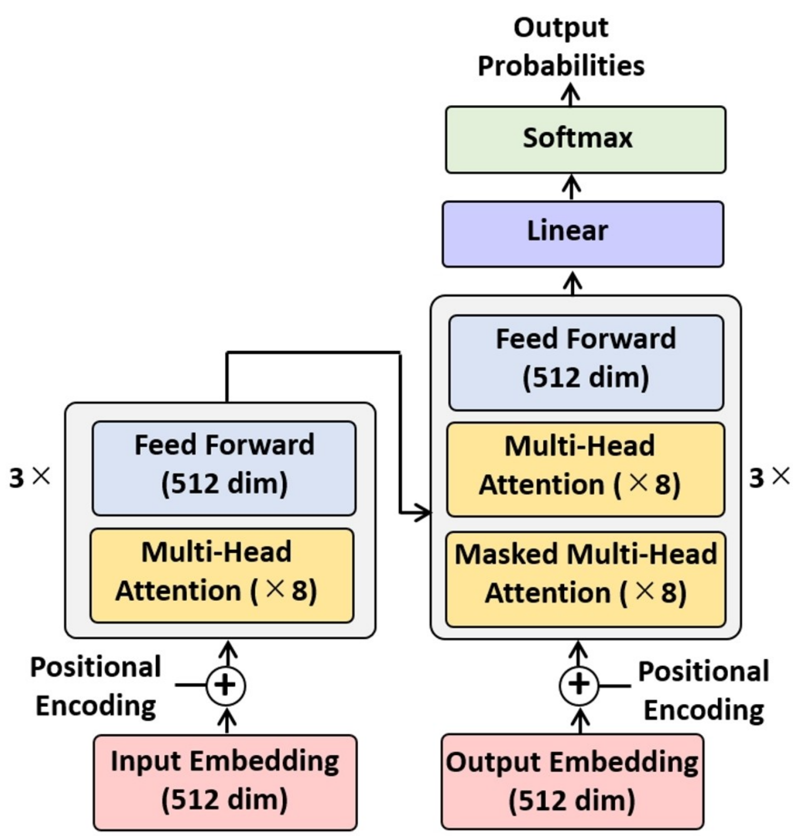

FIGURE 1 Transformer architecture. Shown is a schematic depiction of a representative transformer. "dim" stands for dimensions. The figure was adapted from Ref. [11].

decoder (right) also consists of three modules with multiple attention sub-layers and a feed forward sub-layer. However, different from the encoder, each decoder module contains two types of attention sub-layers (eight each). The multi-head attention sub-layers correspond to those in the encoder and operate on hidden states generated by the encoder and on the output of the first decoder module. These multi-head attention sub-layers learn relationships between sequence encoder and decoder embeddings via the self-attention mechanism. By contrast, the masked attention sub-layers exclusively process the output of the preceding attention sub-layer of the decoder modules. They detect and mask transmitted information that might violate the correct sequential order of translated encodings, thereby preventing translation errors. Output tokens are sampled based on the probability distribution learned by the transformer.

3 | PRIMARY APPLICATION AREAS FOR CHEMICAL LANGUAGE MODELS

The design of virtual compound libraries via machine translation was one of the first applications of CLMs, for which RNNs [2,5,12,13] and, subsequently, transformers were used [13,14]. In addition, transformer networks have been employed for other generative compound design applications [15–17]. Furthermore, both RNN and transformer models have been extensively used for chemical reaction prediction [18–21], representing a growth area for CLM applications. Furthermore, transformer variants have been employed to predict new compounds from protein sequences [11,22–24]. These applications translating protein sequences into molecular strings are closely related to others focusing on the prediction of drug-target interactions [24–26]. In light of recent CLM developments, it is apparent that transformers are beginning to be preferentially used for many applications. A likely reason for this trend is the conceptual and practical attractiveness of self-attention mechanisms (as exemplified by nearly 93,000 current citations of the original transformer reference [6]). However, this does not supersede the relevance of RNN architectures for CLM applications, especially for learning from strictly sequential (time step) data.

For transformer model interpretation, self-attention weights can be represented in attention maps [27,28]. Such maps contain input and output data in columns and rows, respectively, and weights in corresponding color-coded cells. Hence, these maps enable visualization of attention patterns determining machine translations and predictions.

4 | EXEMPLARY CHEMICAL LANGUAGE MODELS

We have developed CLMs for applications in medicinal chemistry. In the presentation at the 8th Autumn School of Chemoinformatics, exemplary CLMs designed for different prediction tasks are discussed.

For example, three LSTM-based RNNs were combined following the molecular hierarchy of the structure-activity relationship (SAR) matrix (SARM) method to further extend its fragment-based design capacity (DeepSARM) [29,30]. SARM generates new compounds by recombining core structure and substituent fragments extracted from related analogue series that are organized in matrices reminiscent of R-group tables. For the generation of novel fragments, three LSTM encoder-decoder units were implemented to act in concert, as illustrated in Figure 2.

Following the compound fragmentation hierarchy of the SARM approach, combination of a Key 2 and Value 2 fragment produces a compound core structure (Key 1 fragment) that is then combined with the Value 1 fragment (substituent) to generate a new compound. The RNN models are pre-trained using large numbers of fragments from compounds with activity against a target class (such as protein kinases) and then fine-tuned using fragments from compounds that are active against a target from that class (an individual kinase). The Key 2, Value 2, and Value 1 Generator networks produce novel fragments that are prioritized for matrix expansion based on (model-internal) log-likelihood scores [29,30]. Combining DeepSARM fragments yields a wealth of new candidate compounds.

For a conceptually distinct application aiming to design new peptidomimetics, the DeepCubist method was developed [31]. In this case, an unconditioned transformer model was derived to convert sp3-rich carbon skeletons mimicking different peptide secondary structure elements into chemically accessible scaffolds containing heteroatoms and bond orders. The transformer was found to retain input skeletons in newly generated peptidomimetic scaffolds at higher rates than encoder-decoder RNNs.

Different from chemical structure conversion via generative modeling, the inclusion of molecular property constraints in compound design requires the implementation of conditional transformers models. For example, such transformers were derived to predict activity cliffs [32] or individual highly potent compounds [33].

During pre-training, these CLMs learned mappings of source to target compounds from many different activity classes conditioned on potency differences

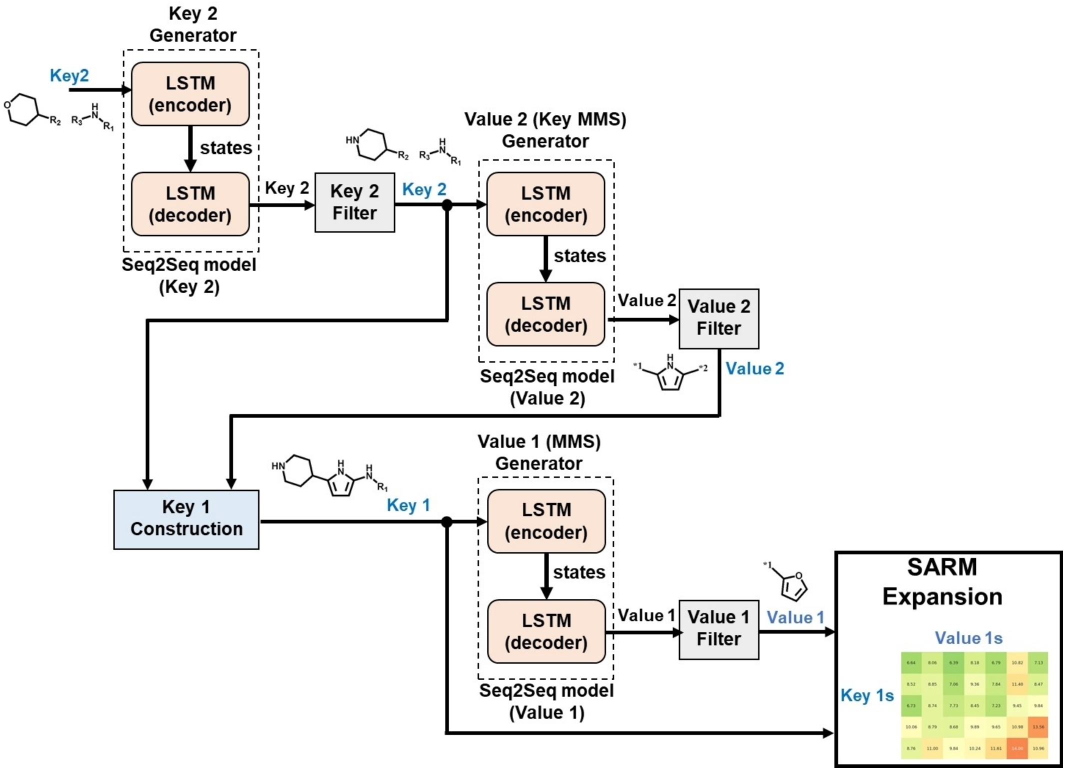

FIGURE 2 DeepSARM. The computational architecture of DeepSARM comprising three encoder-decoder LSTM RNNs ("Generators") is depicted (figure adapted from Ref. [30]).

caused by structural modifications of source compounds:

(Source compound, Potency difference)→(Target compound).

Tokenization was facilitated by transforming SMILES strings into single-character tokens, two-character tokens (e.g. "Br"), and tokens in brackets (e.g. "[nH]"). In addition, the range of potency differences between source and target compounds was partitioned into equally sized bins encoded as single tokens. Each potency difference was assigned to the token of the corresponding bin. The constructed chemical vocabulary consisted of all generated tokens plus "start" and "end" tokens marking the start and end points of a sequence.

A pre-trained CLM was then fine-tuned on compound pairs from individual activity classes. For (*Source compound, Potency difference*) test instances from a given class, the model was used to generate a set of candidate target compounds intended to meet the potency difference constraints and have higher potency than the source compounds.

Following this approach, the DeepAC model predicted activity cliffs with an accuracy at least comparable to top-performing machine learning models and further extended such predictions through generative design of new activity cliff compounds [32]. Furthermore, compound predictions conditioned on large potency differences were generalized beyond activity cliffs by adjusting the training protocol to predict highly potent compounds from weakly potent input templates [33]. The transformer CLM was shown to successfully predict known highly potent compounds from different activity classes not encountered during training. Moreover, the model was capable of creating highly potent compounds that were structurally diverse [33], as shown in Figure 3.

In addition, a CLM variant was implemented to enable generative design of highly potent compounds in low-data regimes based on meta-learning [34]. Using varying amounts of fine-tuning data, the CLM meta-learner consistently generated compounds with higher potency than a corresponding transformer without the meta-learning extension [34].

Another CLM was developed for lead optimization in medicinal chemistry. This model was trained to iteratively extend analogue series with new potent compounds [35]. For this purpose, analogue series can be perceived as sequences of R-groups attached to an invariant core structure, which provides the conceptual basis for series extension via CLMs. Given the strictly sequential nature of the data and predictions, an RNN architecture was implemented comprising an LSTM

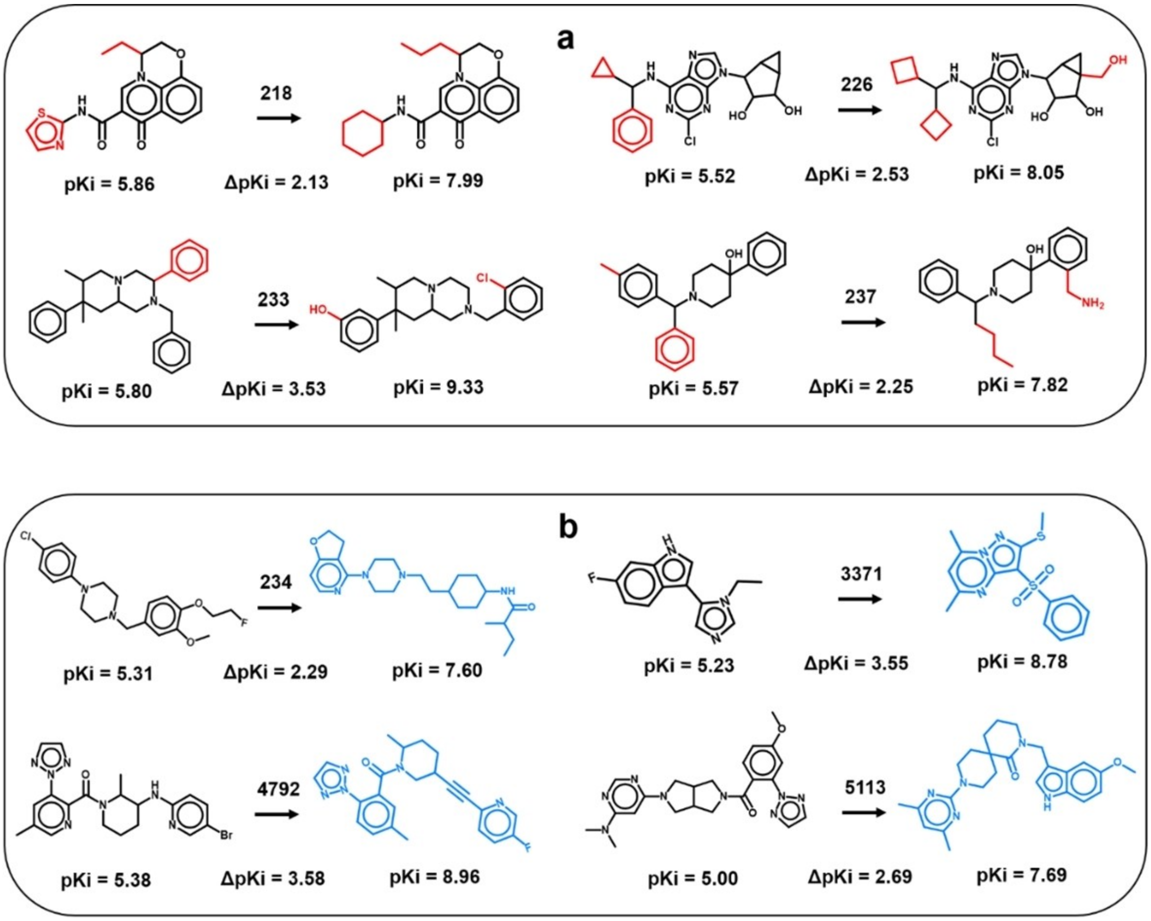

FIGURE 3 Design of potent compounds. Shown are pairs of input compounds (left of each arrow) and new compounds predicted by the CLM (right) including (a) potent structural analogues of input compounds (conserved core: black, distinguishing substituents: red) and (b) potent compounds with distinct core structures and substituents (blue). Numbers on arrows indicate different biological activities. All predicted compounds represented correctly reproduced known compounds never encountered during model derivation. For each compound, its pK_i value is reported and for each pair, the potency difference (figure adapted from Ref. [33]).

layer followed by batch normalization and dense layers. This model (termed DeepAS) was trained on a large collection of nearly 100,000 analogue series with activity against a total of ~2000 targets. In this case, fine-tuning on specific series was not required. All analogue series used for training were arranged in the order of increasing compound potency. Hence, the DeepAS model implicitly learned potency gradients captured by analogue sequences. New R-groups were then predicted by the dense layer of the model based on conditional probabilities derived from preceding R-groups. Given the ordering of training series, iterative extension of analogue series using DeepAS was implicitly directed towards compounds with increasing potency, and the model correctly reproduced potent analogues for many test series [35].

Finally, a transformer was also implemented to predict new active compounds from sequence motifs of ligand binding sites (*vide supra*) [11]. Hence, in this case, protein sequences were mapped to compound strings. In a proof-of-concept application, the Motif2 Mol model was derived using ATP site-directed inhibitors of more than 200 human kinases and extended sequence motifs encompassing their ATP binding sites. Motif2 Mol consistently reproduced known inhibitors of test kinases not encountered during training [11].

5 | CONCLUSIONS

Language models are becoming increasingly popular for machine translation tasks in many areas and scientific fields including chemistry and drug discovery. A major attraction of CLMs is their versatility in relating different types of sequence representations to each other, hence opening the door to exploring new applications that were difficult to address before. While RNNs including RNN-based VAEs have been widely applied for processing of sequential (time step) data, transformer CLMs are beginning to dominate the field. This is largely due to their integral self-attention mechanism, which reduces errors in generated vectors and increases computational efficiency through parallel processing, and, in addition, to the relative ease with which transformers can be conditioned on molecular properties or other constraints. However, these transformer features do not render encoder-decoder RNNs (or other RNN architectures) obsolete. Depending on the applications and underlying data structures, these RNNs yield effective models, for example, as discussed herein for applications in fragment-based compound design (DeepSARM). Generative modeling and chemical reaction prediction have been main application areas for CLMs thus far. For unconditioned generative compound design, the DeepCubist approach provides an exemplary application. Furthermore, the prediction of new compounds from protein sequence information has been attempted, given the capacity of language models to translate different types of sequence data. These studies are closely related to predictions of drug-target interactions using CLMs. In compound design, specialized applications can be conceived by conditioning CLMs on specific (property) constraints, with DeepAC and generalized prediction of highly potent compounds being representative examples. Molecular design with specific constraints is expected to be another growth area for CLMs, perhaps also including concepts from geometric deep learning for structure-based design. Finally, a strength of CLMs is that they can be deployed with any compound or target data that can be organized in sequential form and appropriately tokenized, as discussed herein for the CLM-based extension of analogue series with newly designed compounds. In this case, a relatively simple RNN variant was sufficient to obtain accurate predictions. Clearly, data transformation into sequences will provide a basis for off-the-beaten path applications of CLMs in drug discovery and design that have not been considered before.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available in ChEMBL at https://www.ebi.ac.uk/chembl/. These data were derived from the following resources available in the public domain: – Compound data, https://www.ebi.ac.uk/chembl/.

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